FINAL PROJECT REPORT

Internship Project Report — Project Initialization, Planning, Data Analysis, Visualization and Reporting

Global AI in Education Analysis using Tableau

Submitted by

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Submission Date: 30 July 2026

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1. Introduction

1.1 Project Overview

Artificial Intelligence (AI) has moved from being an experimental technology to a central force reshaping how education is delivered, managed, and experienced around the world. Adaptive learning platforms, AI-powered tutoring systems, automated grading tools, and intelligent classroom-management applications are increasingly present in schools and universities across every continent. Yet the pace of this transformation is far from uniform. Countries differ substantially in how quickly their schools adopt AI, in the digital infrastructure that supports it, in the strength of government policy governing its use, and in the extent to which urban and rural learners benefit equally. For policymakers, educators, and institutions trying to plan for this change, the underlying data is often scattered across disconnected indicators, making it difficult to see the global picture clearly.

This is precisely the gap that modern data visualization is designed to close. Raw numbers on AI adoption, internet penetration, education indices, and usage patterns are of limited value on their own; their significance only becomes clear when they are organised, compared, and presented in a form a decision-maker can interpret at a glance. Visualization converts abstract statistics into recognisable patterns, such as a rising trend line, a shaded map, or a cluster of points on a scatter plot, that communicate insight far more efficiently than a spreadsheet of figures ever could. Among the tools available for this purpose, Tableau stands out for its ability to connect directly to a dataset and produce interactive, publication-quality dashboards and stories without requiring extensive programming knowledge, making sophisticated analysis accessible to educators and administrators as well as data professionals.

The purpose of this project, Global AI in Education Analysis using Tableau, is to bring exactly this kind of clarity to the subject of AI adoption in education. Using a global dataset that tracks indicators such as school, student, and teacher AI usage, internet penetration, education index, and government AI policy status across ten countries and six world regions between 2015 and 2026, the project builds a set of Key Performance Indicators, interactive dashboards, and a guided Tableau Story that together allow a viewer to explore how AI adoption has evolved over time, how it varies geographically, and what factors appear to accompany its growth. This particular dataset was selected because it combines exactly the dimensions, time, geography, infrastructure, and policy, that are needed to answer meaningful questions about digital transformation in education, rather than presenting AI adoption as an isolated statistic.

The resulting dashboards and story are intended to be useful well beyond the scope of a single project. Governments and policymakers can use the geographic and policy-distribution views to identify where regulatory support is lagging behind adoption. Researchers can use the Education Index and internet-penetration comparisons to investigate what enables successful AI integration. School and university administrators can benchmark their own progress against regional and global trends. In each case, the underlying goal is the same: to support business intelligence and data-driven decision-making, replacing intuition and anecdote with a shared, interactive view of the evidence.

1.2 Objectives

1. To analyse the global adoption of Artificial Intelligence in education systems across multiple countries and regions between 2015 and 2026.

2. To compare AI implementation across countries, including the United States, United Kingdom, Germany, Canada, Australia, China, India, Brazil, Nigeria, and Pakistan.

3. To build meaningful Key Performance Indicators that summarise AI adoption and usage across schools, students, and teachers.

4. To design interactive Tableau dashboards that consolidate AI adoption trends, digital infrastructure, and policy readiness into a single analytical view.

5. To create a Tableau Story that presents a sequential, narrative-driven analysis of global AI adoption in education.

6. To compare regional trends in AI adoption and digital infrastructure across Africa, Asia, Europe, North America, Oceania, and South America.

7. To analyse government AI policy readiness and its distribution across Active, Draft, and No-Policy categories.

8. To support data-driven decision-making for educators, administrators, researchers, and policymakers through interactive visual analytics.

9. To improve understanding of the relationship between a country's Education Index and its level of school AI adoption.

10. To examine the digital divide between urban and rural AI usage and highlight areas that may require policy intervention.

SMARTBRIDGE
Let's Bridge the Gap

2. Project Initialization and Planning Phase

Date	30 July ,2026
Team ID	Shrishti Jaiswal
Project Name	Global AI in Education Analysis using Tableau
Maximum Marks	3 Marks

2.1 Define Problem Statements (Customer Problem Statement Template):

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS- 1	Education Policy Maker	Analyze global AI adoption in education systems	It is difficult to compare AI adoption, investment, and outcomes across countries	Educational AI data is scattered across multiple indicators and lacks a unified dashboard	Unable to make informed policy decisions efficiently
PS- 2	School/University Administrator	Monitor AI implementation and its impact on student learning	It is difficult to identify trends and evaluate AI effectiveness	There is no interactive platform that combines key education and AI metrics in one place	Challenged in making data-driven decisions to improve educational outcomes

2. Project Initialization and Planning Phase

Date	30 July 2026
Team ID	Shrishti Jaiswal
Project Title	Global AI in Education Analysis using Tableau
Maximum Marks	3 Marks

2.2 Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To analyze global AI adoption in education using Tableau and develop interactive dashboards, KPIs, and a story for data-driven decision-making.
Scope	Includes data collection, cleaning, KPI creation, charts, dashboards, Tableau Story, and workbook publishing. Excludes predictive modelling and real-time data integration.
Problem Statement	
Description	AI in education data is distributed across multiple indicators, making comparison and analysis difficult using raw datasets.
Impact	The solution provides a centralized interactive dashboard that helps educators, administrators, researchers, and policy makers make informed decisions.
Proposed Solution	

Approach	Collect, clean, and transform the dataset in Tableau, then create KPIs, visualizations, dashboards, and a Tableau Story with interactive filters.
Key Features	Interactive KPIs, dynamic dashboards, Tableau Story, country filters, trend analysis, and easy-to-understand visualizations.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Laptop/Desktop (Intel i3 or above)		
Computing Resources	Local Development Engine	Minimum 4 GB RAM
Memory	Local RAM	500 MB free space
Storage	Disk space for data, models, and logs	Max 1 GB
Tableau Public / Tableau Desktop		
Frameworks	Supporting Tool	Microsoft Excel (CSV preparation)
Libraries	Development Environment	Tableau Public and Microsoft Excel
Development Environment	IDE, version control	VS Code (or Text Editor), macOS Terminal, Git, and GitHub UI
CSV format (cleaned dataset for Tableau analysis)		
Data	Global AI in Education Dataset	CSV format (cleaned dataset for Tableau analysis)

2.3 Initial Project Planning Template

Date	30 July 2026
Team ID	Shrishti Jaiswal
Project Name	Global AI in Education Analysis using Tableau
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection	USN-1	As a user, I want to collect global AI in Education data from reliable sources.	2	High	1	16 July 2026	20 Jul 2026
Sprint-1	Data Preprocessing	USN-2	As a user, I want to clean and prepare the dataset for accurate analysis.	2	High	1	16 July 2026	20 Jul 2026
Sprint-2	Visualization	USN-3	As a user, I want to create KPI cards and interactive visualizations in Tableau.	3	High	1	21 Jul 2026	25 Jul 2026
Sprint-2	Dashboard	USN-4	As a user, I want to build dashboards and a story to present AI education insights.	3	Medium	1	21 Jul 2026	25 Jul 2026
Sprint-3	Reporting	USN-5	As a user, I want to publish the Tableau workbook and prepare the final project report.	2	High	1	26 Jul 2026	30 Jul 2026

3. **Data Collection and Preprocessing Phase**

Date	30 July 2026
Team ID	Shrishti Jaiswal
Project Title	Global AI in Education Analysis using Tableau
Maximum Marks	2 Marks

3.1 Data Collection Plan & Raw Data Sources Identification Template

Elevate your data strategy with the Data Collection plan and the Raw Data Sources report, ensuring meticulous data curation and integrity for informed decision-making in every analysis and decision-making endeavor.

Data Collection Plan Template

Section	**Description**
Project Overview	The Global AI in Education Analysis using Tableau project analyzes worldwide AI adoption in education using an interactive Tableau dashboard and story. It presents KPIs and visualizations to compare AI adoption, investment, infrastructure, institutional readiness and student engagement across countries.
Data Collection Plan	The project uses a Global AI in Education CSV dataset collected from public sources. The dataset was cleaned by removing duplicates, handling missing values, standardizing country and category names, and validating data types before importing into Tableau.

Raw Data Sources Identified	1. Global AI in Education Dataset containing country-wise AI adoption indicators. 2. Public education statistics and AI implementation indicators used for visualization and comparative analysis.

Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	Global AI in Education Dataset	https://drive.google.com/file/d/1m-BpmDp9oWII4NeIzuKlg__axJZZJTuW/view?usp=sharing	CSV	Approx. 300 KB	Public

3. Data Collection and Preprocessing Phase

Date	30 July 2026
Team ID	Shrishti Jaiswal
Project Title	Global AI in Education Analysis using Tableau
Maximum Marks	3 Marks

3.2 Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	**Data Quality Issue**	**Severity**	**Resolution Plan**
Global AI in Education Dataset	Missing values in selected records, duplicate entries, inconsistent country and institution names, inconsistent text formatting, blank fields and incorrect data types that may affect Tableau dashboards and story analysis.	Moderate	Removed duplicate records, handled missing values, standardized country and institution names, corrected text formatting, validated data types, and verified the cleaned dataset before creating KPIs, visualizations, dashboards and Tableau Story.

3.Data Collection and Preprocessing Phase

Date	30 July 2026
Team ID	Shrishti Jaiswal
Project Title	Global AI in Education Analysis using Tableau
Maximum Marks	10 Marks

3.3 Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	**Description**
Data Overview	The dataset contains global AI in Education data including country, institution type, AI adoption, investment, digital infrastructure, student engagement, institutional readiness and education indicators.
Data Cleaning	Missing values handled, duplicates removed, formatting standardized and inconsistent values corrected before Tableau analysis.
Data Transformation	Created calculated fields, KPIs, aggregations and filters to prepare data for Tableau dashboards and stories.
Data Type Conversion	Validated numeric, categorical and date fields with appropriate data types for Tableau.
Column Splitting and Merging	Split and merged relevant columns for better filtering and analysis.
Data Modeling	Structured the dataset for Tableau relationships, dashboards, KPIs and story creation.

Save Processed Data	Saved the processed dataset in CSV/Excel format and imported it into Tableau.

4. Data Visualization

4.1 Business Question and Visualization Report

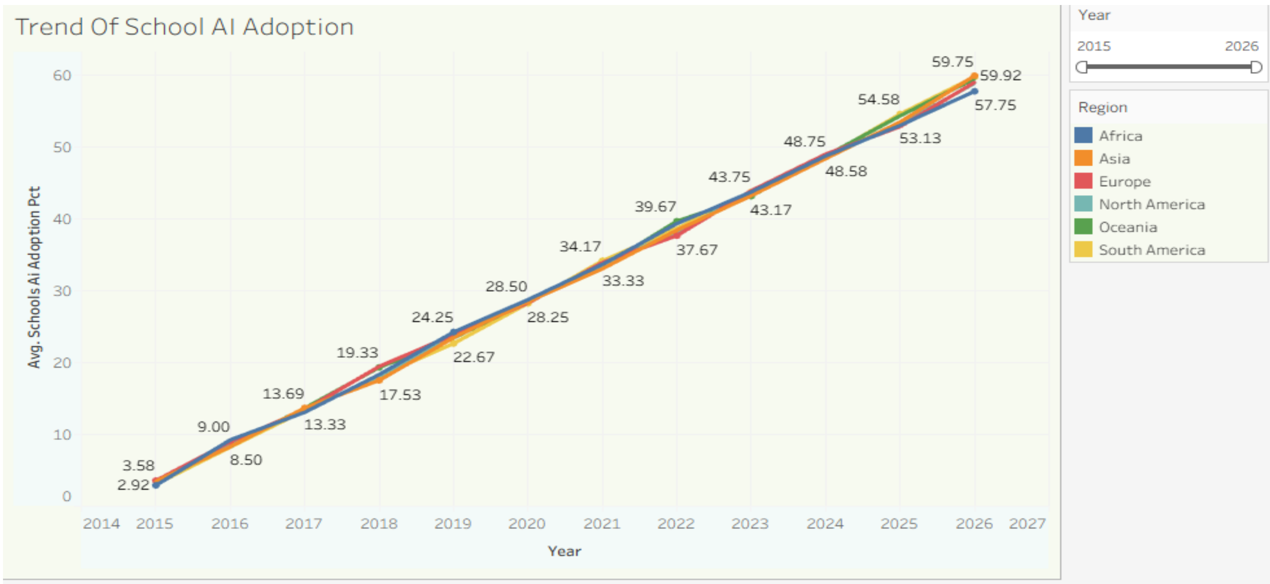
Date	30 July 2026
Team ID	Shrishti Jaiswal
Project Name	Global AI in Education Analysis using Tableau
Maximum Marks	5 Marks

Business Questions and Visualizations

Q1. How has school AI adoption changed across regions from 2015–2026?

Visualization: A multi-line chart illustrating the average School AI Adoption Percentage across Africa, Asia, Europe, North America, Oceania, and South America from 2015 to 2026.

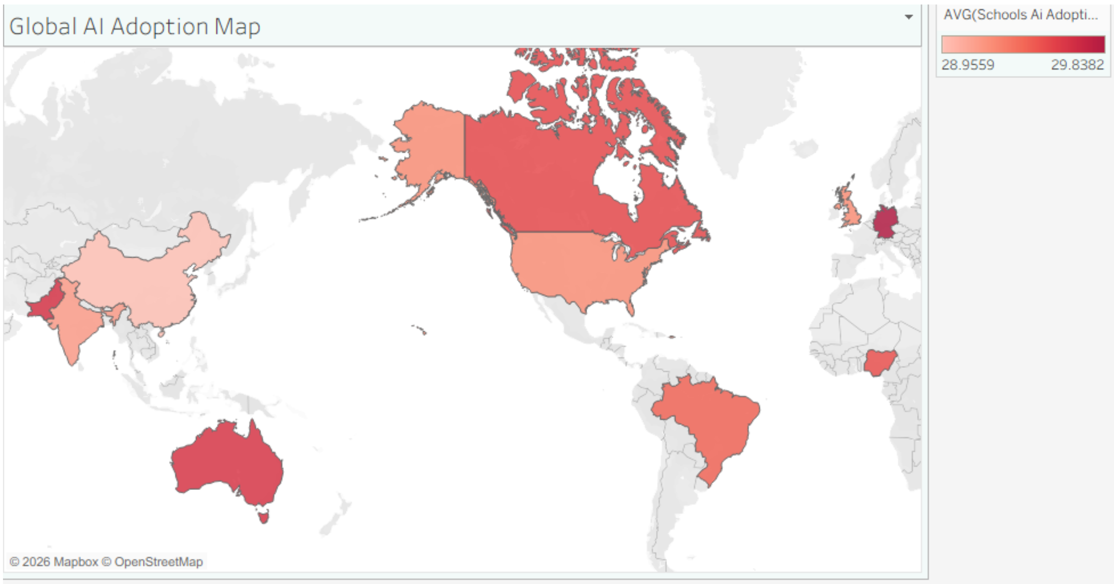
Insight Provided: The visualization demonstrates a consistent increase in school AI adoption across all regions over the years. While every region shows positive growth, slight variations indicate different adoption rates influenced by technological infrastructure and educational policies.



Q2. Which countries have the highest average School AI Adoption Percentage?

Visualization: A filled world map representing the average School AI Adoption Percentage for each country using color intensity.

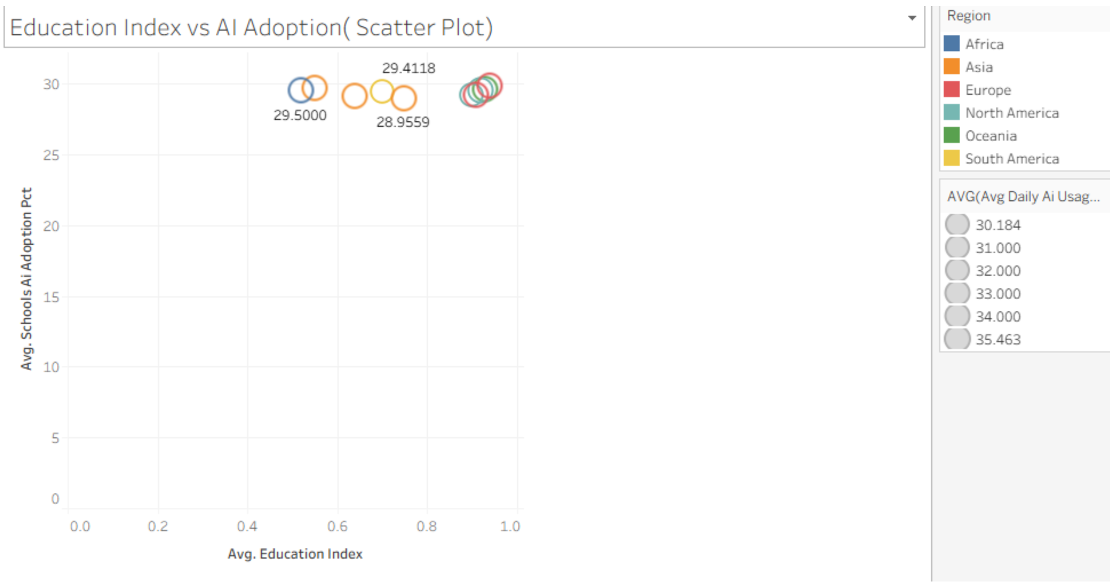
Insight Provided: The map highlights geographical differences in AI adoption. Countries with darker shades indicate higher AI implementation in education, helping identify leaders and regions requiring additional digital transformation.



Q3. Is there a relationship between Education Index and AI adoption?

Visualization: A scatter plot comparing Education Index on the X-axis and School AI Adoption Percentage on the Y-axis, with bubble size representing Average Daily AI Usage.

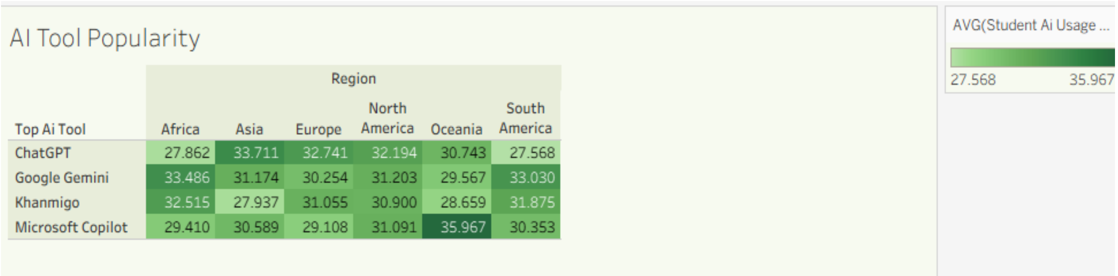
Insight Provided: The scatter plot suggests a positive relationship between Education Index and AI adoption. Countries with stronger educational development generally exhibit higher AI integration in schools.



Q4. Which AI tools are most popular across regions?

Visualization: A highlight table comparing the popularity of ChatGPT, Google Gemini, Khanmigo, and Microsoft Copilot across different regions.

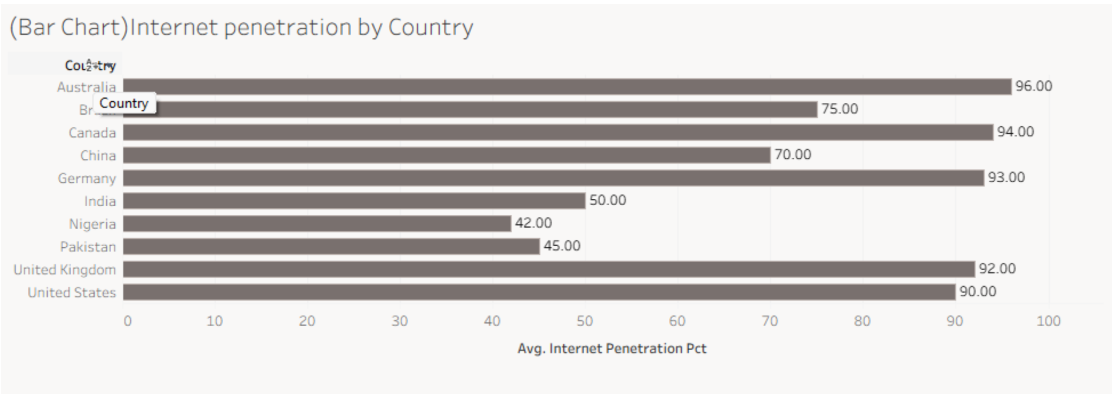
Insight Provided: The visualization compares AI tool usage across continents, helping identify regional preferences and showing which AI applications are widely adopted for educational purposes.



Q5. How does internet penetration vary by country?

Visualization: A horizontal bar chart showing the average Internet Penetration Percentage for each country.

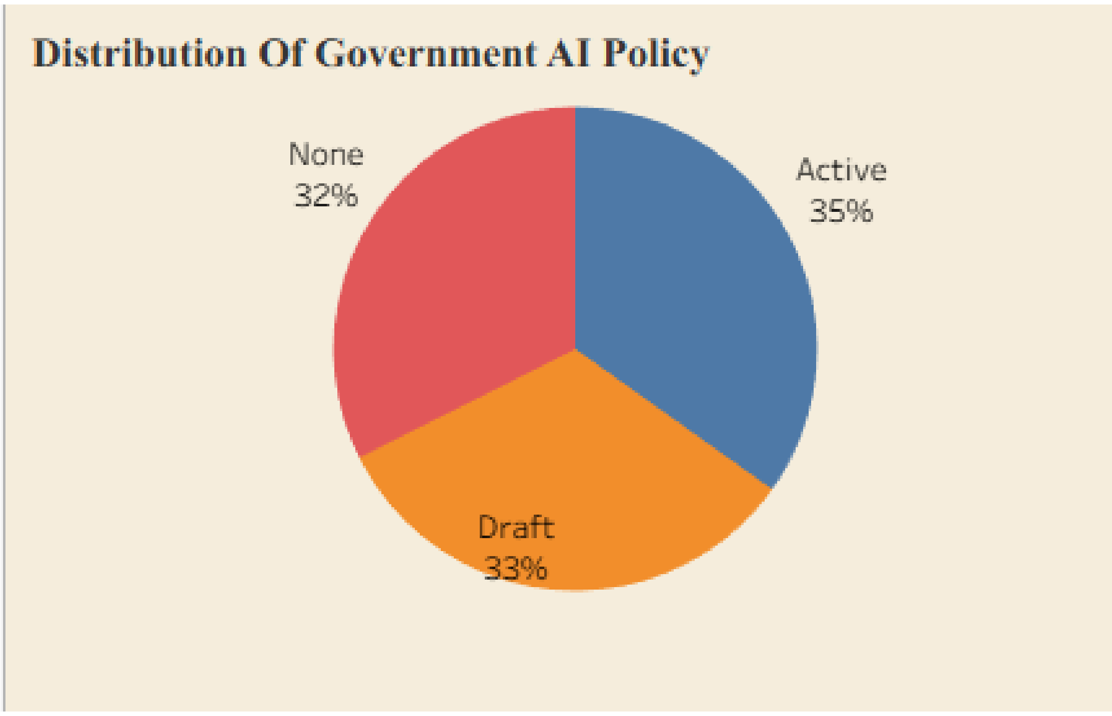
Insight Provided: Countries with higher internet penetration provide stronger digital infrastructure, creating a favorable environment for AI adoption in education and online learning.



Q6. What is the current distribution of Government AI Policy among countries?

Visualization: A pie chart showing the proportion of countries with Active, Draft, and No Government AI Policy.

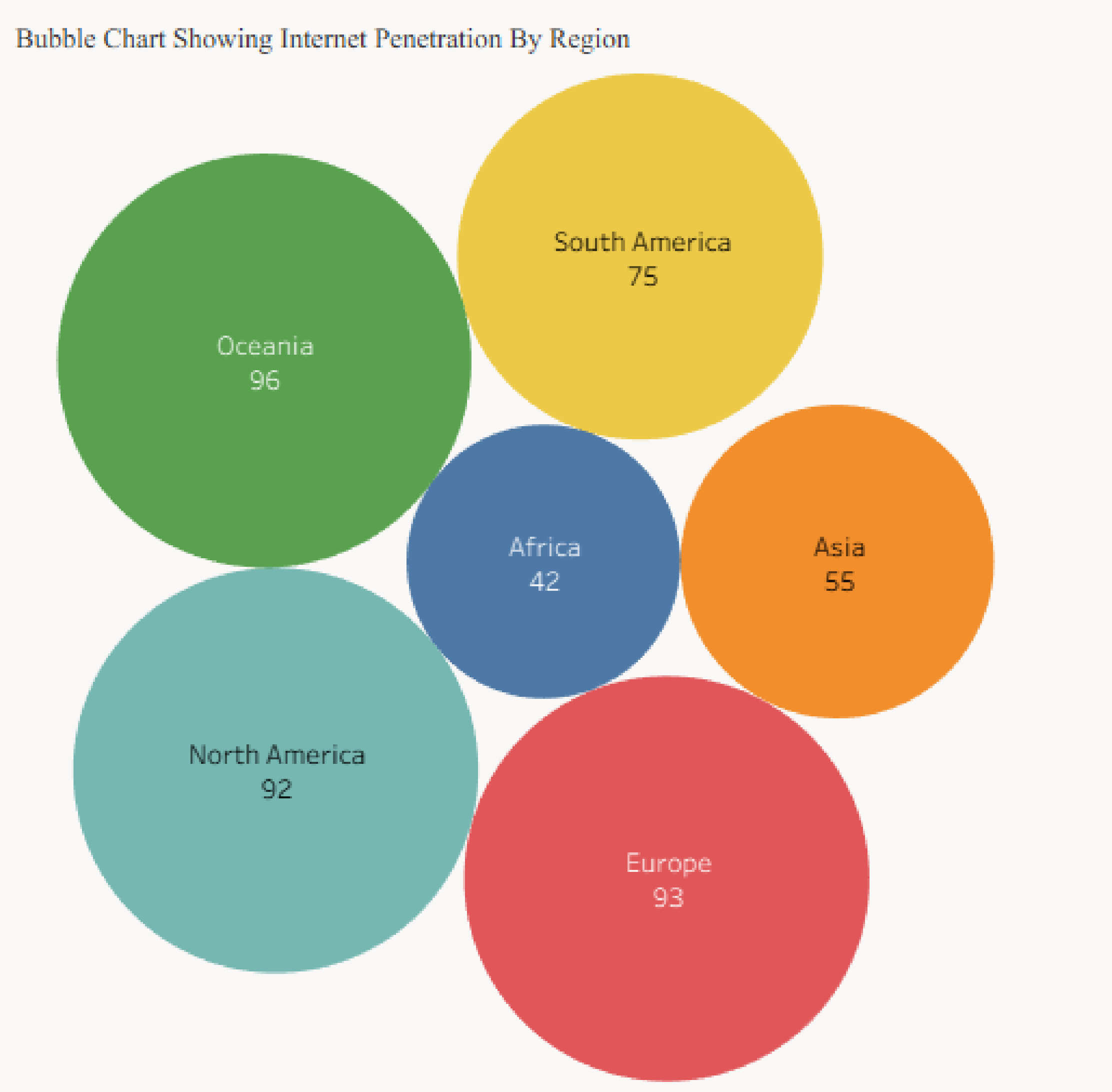
Insight Provided: The visualization presents the policy readiness of different countries for AI adoption. A higher proportion of active policies indicates stronger governmental support for AI implementation in education.



Q7. How does internet penetration compare across different region?

Visualization: A packed bubble chart representing average Internet Penetration Percentage across different geographical regions.

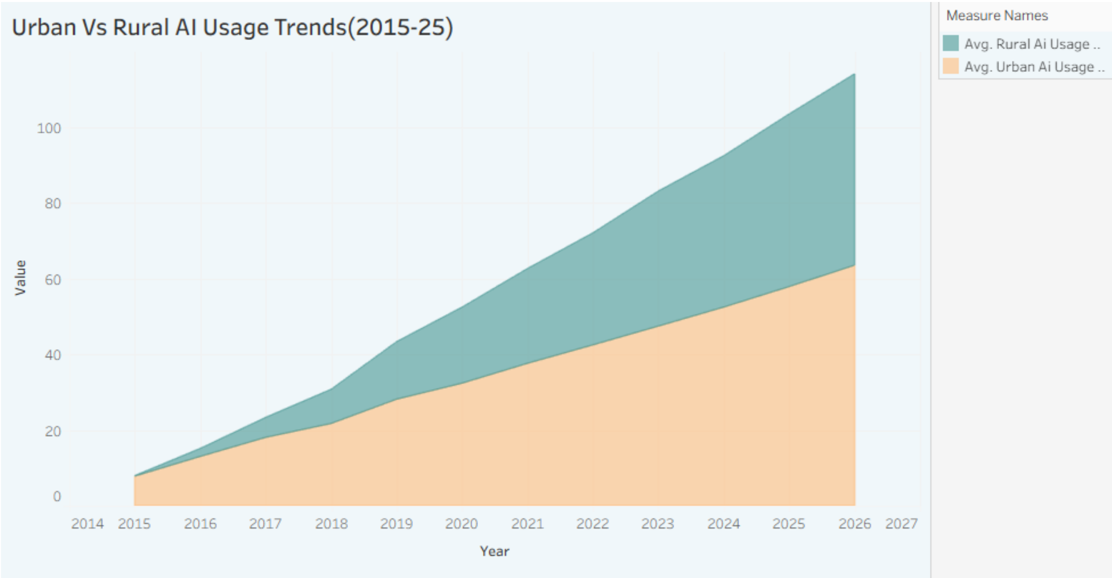
Insight Provided: The bubble chart provides an easy comparison of internet accessibility among regions. Better internet connectivity supports wider AI adoption and digital education initiatives.



Q8. How have urban and rural AI usage trends evolved?

Visualization: An area chart comparing Urban AI Usage Percentage and Rural AI Usage Percentage from 2015 to 2026.

Insight Provided: The visualization indicates continuous growth in AI usage in both urban and rural areas. Urban regions consistently maintain higher AI usage, highlighting the ongoing digital divide while showing gradual improvement in rural adoption.



Q9. What key performance indicators summarize AI adoption?

Visualization: A KPI dashboard summarizing Average Daily AI Usage, Gender AI Adoption, Rural AI Usage, School AI Adoption, Student AI Usage, Teacher AI Usage, and Urban AI Usage.

Insight Provided: The KPI dashboard provides a concise overview of the key metrics in the dataset. It enables stakeholders to quickly assess AI adoption levels among schools, students, teachers, and urban-rural populations, supporting informed educational decision-making.

KPI Sheet

Avg. Avg Daily Ai ..	Avg. Gender ..	Avg. Rural Ai..	Avg. Schools..	Avg. Student..	Avg. Teacher..	Avg. Urban A..
33 mins	5 %	22 %	29 %	31 %	30 %	34 %

4.2 Developing Visualizations

Once the business questions were clearly framed, the next step was to translate each question into a visualization capable of answering it efficiently and accurately. Tableau's chart library was used deliberately, with each visualization type selected to match the analytical nature of its underlying question rather than chosen for decorative purposes.

KPI cards were used to summarise the headline metrics, Average Daily AI Usage, School, Student, and Teacher AI Usage, Urban and Rural AI Usage, and the Gender Gap in AI Usage, giving viewers an immediate executive-level snapshot before they explore the detailed charts. A multi-line chart traces School AI Adoption Percentage across all six regions from 2015 to 2026, making the shared upward trend easy to compare at a glance, while an area chart performs a similar role for Urban versus Rural AI usage, visually emphasising the widening gap between the two. A scatter plot, with bubble size mapped to Average Daily AI Usage, tests the relationship between Education Index and School AI Adoption, and a filled map extends this geographic comparison to the country level.

Categorical and proportional questions were matched to correspondingly categorical charts: a pie chart for the three-way split of Government AI Policy status, a packed bubble chart for comparing internet penetration across regions, a highlight table for comparing the popularity of four AI tools across six regions, and a horizontal bar chart for ranking ten countries by internet penetration. Finally, these individual visualizations were drawn together into two dashboards and a Tableau Story, so that instead of reading nine independent charts, a viewer can follow one coherent, guided narrative about the state of AI in global education.

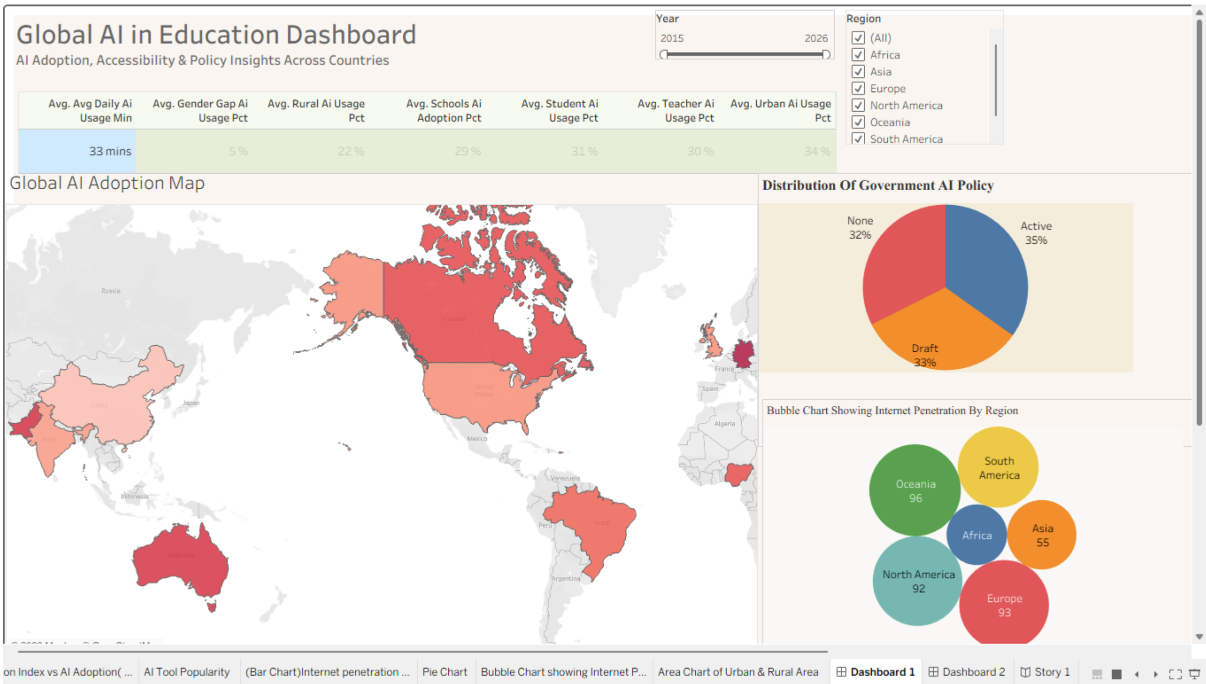
5. Dashboard

5.1 Dashboard Design

Date	30 July 2026
Team ID	Shrishti Jaiswal
Project Name	Global Education
Maximum Marks	5 Marks

Activity 1: Interactive and Visually Appealing Dashboards

Dashboard 1:



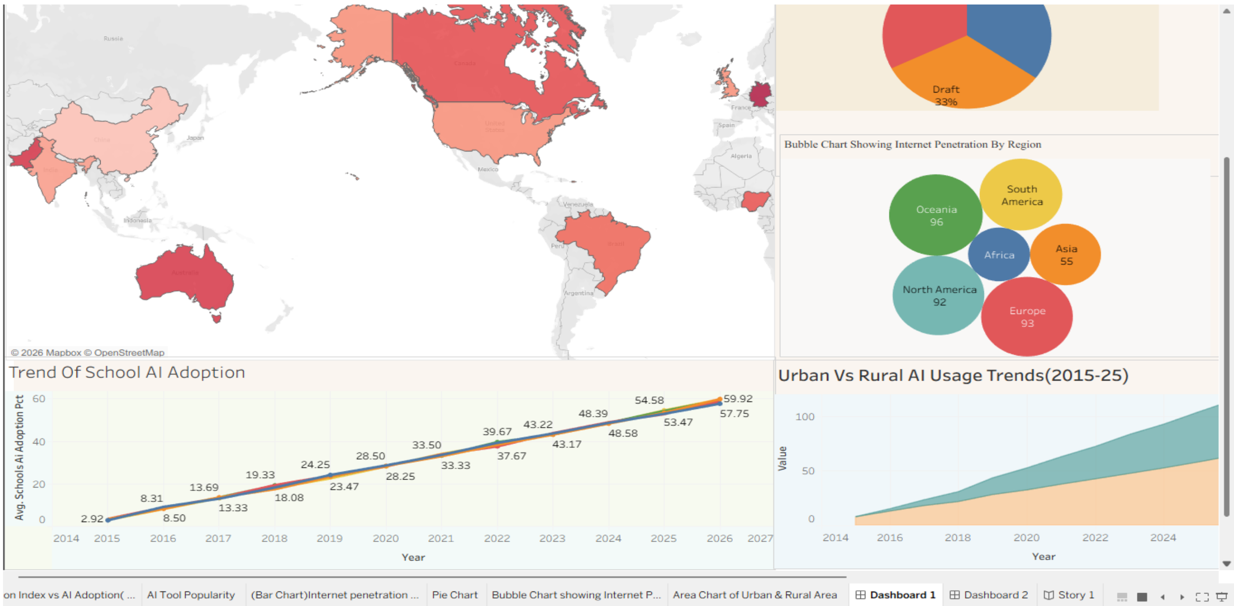
The core dashboard design principles : including a clear and intuitive layout, consistent colour palette, KPI-driven storytelling, interactive filters, geographical analytics, and responsive visual design have been implemented to develop the Global AI in Education Analysis Dashboard. The dashboard transforms raw educational AI data into meaningful business intelligence that supports researchers, educational institutions, and policymakers in evaluating AI adoption trends, digital readiness, and regional educational development. The following analytical insights are generated from the dashboard visualisations:

• Gl**obal AI Adoption Analysis**: The Filled Map provides a comprehensive geographical representation of School AI Adoption Percentage across multiple countries and regions. By using gradient colour encoding, the visual quickly identifies countries demonstrating higher implementation of AI technologies while simultaneously highlighting regions where adoption remains comparatively low. This geographical comparison assists governments and educational organisations in recognising regional disparities and prioritising future AI investments.

• **Comprehensive KPI Performance Summary:** The KPI cards consolidate important educational indicators including Average Daily AI Usage, School AI Adoption, Student AI Usage, Teacher AI Usage, Urban AI Usage, Rural AI Usage, and Gender Gap in AI Usage. These indicators provide stakeholders with an executive-level overview of the present state of AI adoption and enable rapid performance monitoring without analysing multiple individual visualisations.

• **Government AI Policy Readiness:** The Government AI Policy Pie Chart categorises countries into Active Policy, Draft Policy and No Policy groups. The visual demonstrates how policy development directly influences AI implementation within educational systems and helps decision-makers identify countries requiring stronger regulatory frameworks to accelerate responsible AI adoption.

• **Digital Infrastructure Assessment**: The Internet Penetration Bubble Chart compares internet accessibility across geographical regions. The dashboard clearly demonstrates that regions possessing stronger digital infrastructure generally exhibit higher readiness for AI-enabled education, online learning platforms and intelligent educational technologies, thereby emphasising the importance of internet accessibility in successful AI implementation.



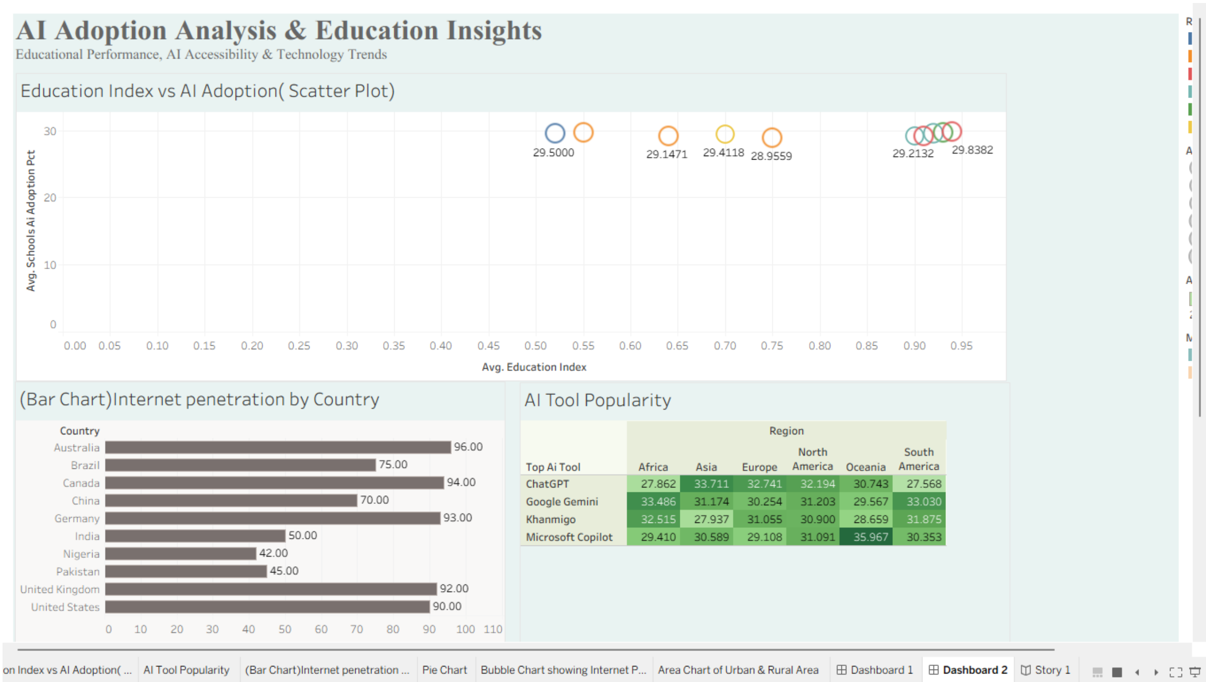
• **School AI Adoption Trend Analysis:** The multi-line trend chart illustrates the growth of School AI Adoption Percentage from 2015 to 2026 across different regions. The continuous upward trend indicates increasing institutional acceptance of AI-powered educational tools and reflects the growing integration of intelligent technologies into teaching, assessment and classroom management processes.

• **Urban and Rural AI Usage Comparison**: The area chart evaluates AI adoption between urban and rural educational environments. Although both categories display positive growth throughout the observed period, urban institutions consistently demonstrate higher adoption levels due to comparatively better infrastructure, internet connectivity and digital resource availability. The visual highlights the continuing digital divide requiring focused policy intervention.

Dashboard 2: Advanced Analytical Insights

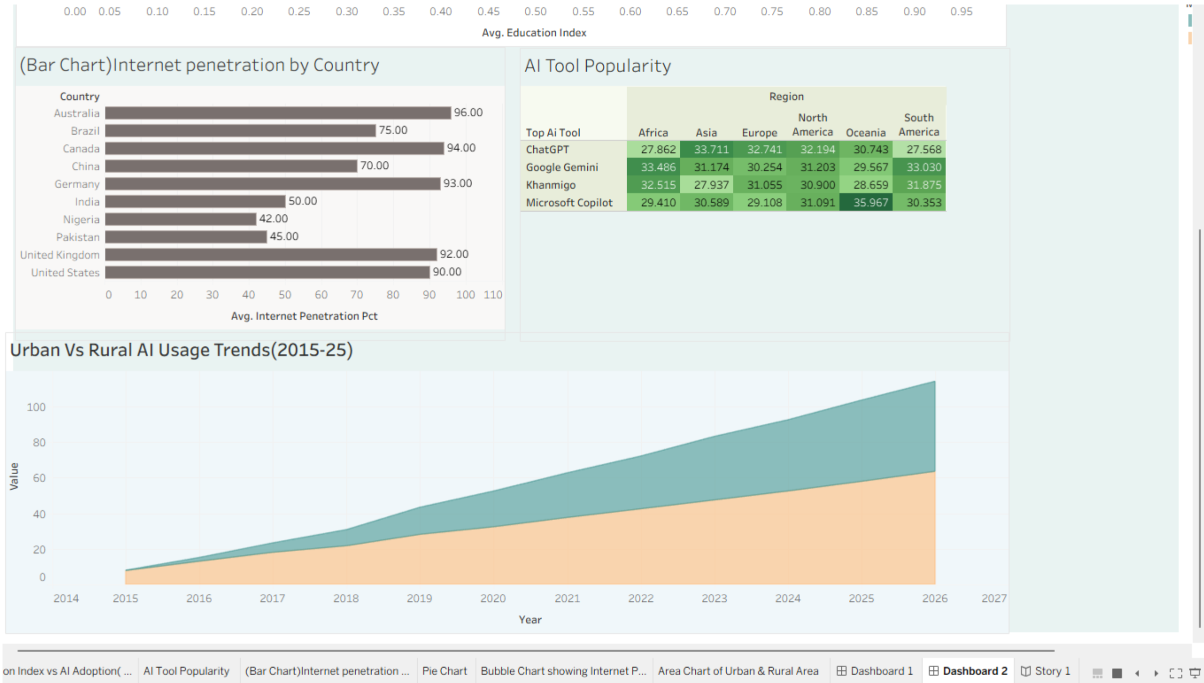
Dashboard 2 provides deeper analytical insights by examining the relationship between educational development, digital infrastructure, and the adoption of Artificial Intelligence across different countries and regions. Unlike Dashboard 1, which presents an overall overview of AI adoption, this dashboard focuses on identifying the factors that influence successful AI implementation in education. The visualizations enable policymakers, educational institutions, and researchers to compare countries, evaluate technological readiness, and identify opportunities for improving AI integration within educational systems.

• **Education Index and School AI Adoption Analysis :** The **Education Index vs School AI Adoption Scatter Plot** evaluates the relationship between a country's educational development and its level of AI adoption in schools. The visualization indicates that countries with stronger education systems generally demonstrate higher AI adoption percentages. This positive correlation suggests that improvements in educational quality, infrastructure, and digital literacy create favourable conditions for implementing AI-powered teaching and learning solutions. The analysis helps identify countries that may require additional investment in education before large-scale AI adoption can be achieved.



• **Internet Penetration by Country :**The **Internet Penetration Bar Chart** compares internet accessibility across different countries. Since AI-powered education depends heavily on reliable internet connectivity, this visualization highlights the importance of digital infrastructure in supporting technology-enabled learning. Countries with higher internet penetration are better positioned to implement online learning platforms, intelligent tutoring systems, and AI-assisted educational tools. The chart also identifies regions where infrastructure development remains essential for expanding AI adoption**.**

• **AI Tool Popularity Across Regions :**The **AI Tool Popularity Heatmap** illustrates the popularity of widely used AI platforms such as **ChatGPT, Google Gemini, Microsoft Copilot, and Khanmigo** across various geographical regions. The visualization reveals regional preferences for different AI tools and demonstrates how educational institutions adopt different technologies based on accessibility, institutional requirements, and user familiarity. This analysis provides valuable insights into current AI usage trends and future opportunities for educational technology providers.



Activity 2: Live Deployment Link

Dashboard 1:

https://public.tableau.com/views/GlobalAIinEducationAnalysisusingTableau/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dashboard 2:

https://public.tableau.com/views/GlobalAIinEducationAnalysisusingTableau/Dashboard2?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Tableau Public Dashboard Link:

Dashboard 1:

https://public.tableau.com/views/GlobalAIinEducationAnalysisusingTableau/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dashboard 2:
https://public.tableau.com/views/GlobalAIinEducationAnalysisusingTableau/Dashboard2?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

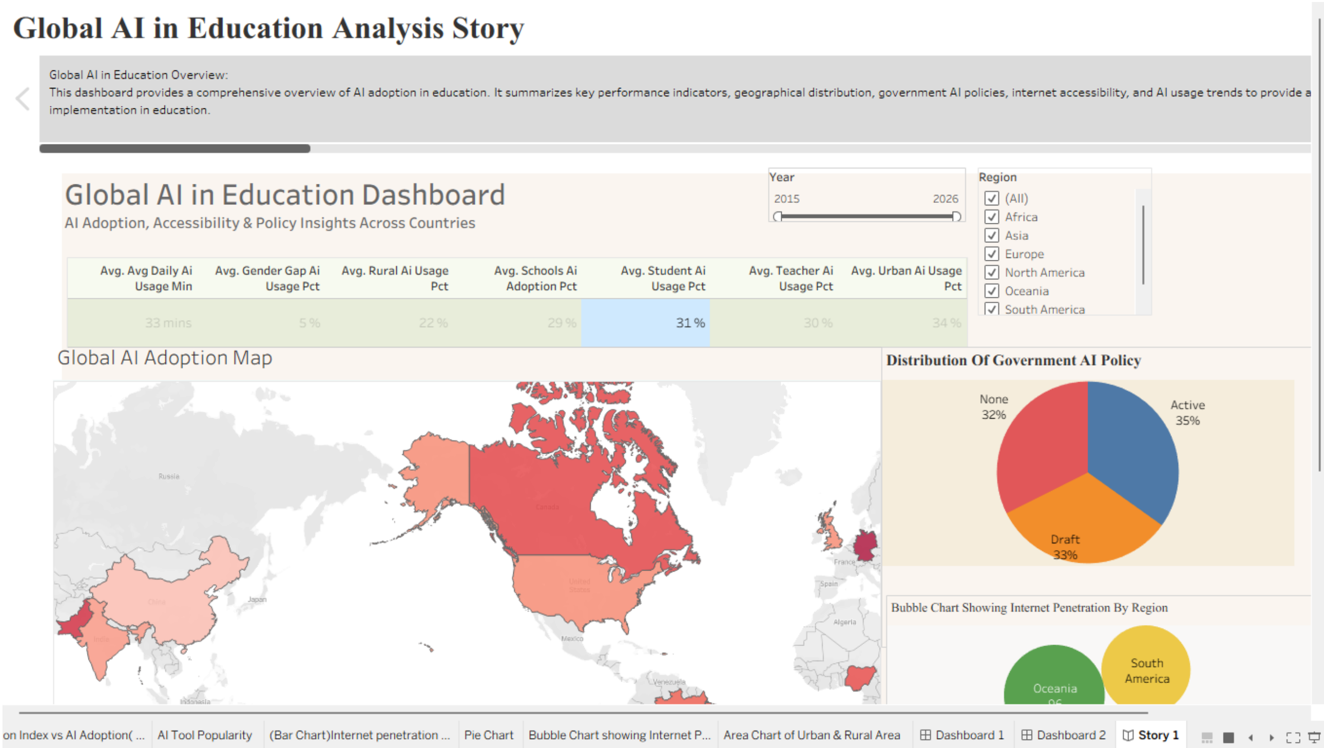
6. Report

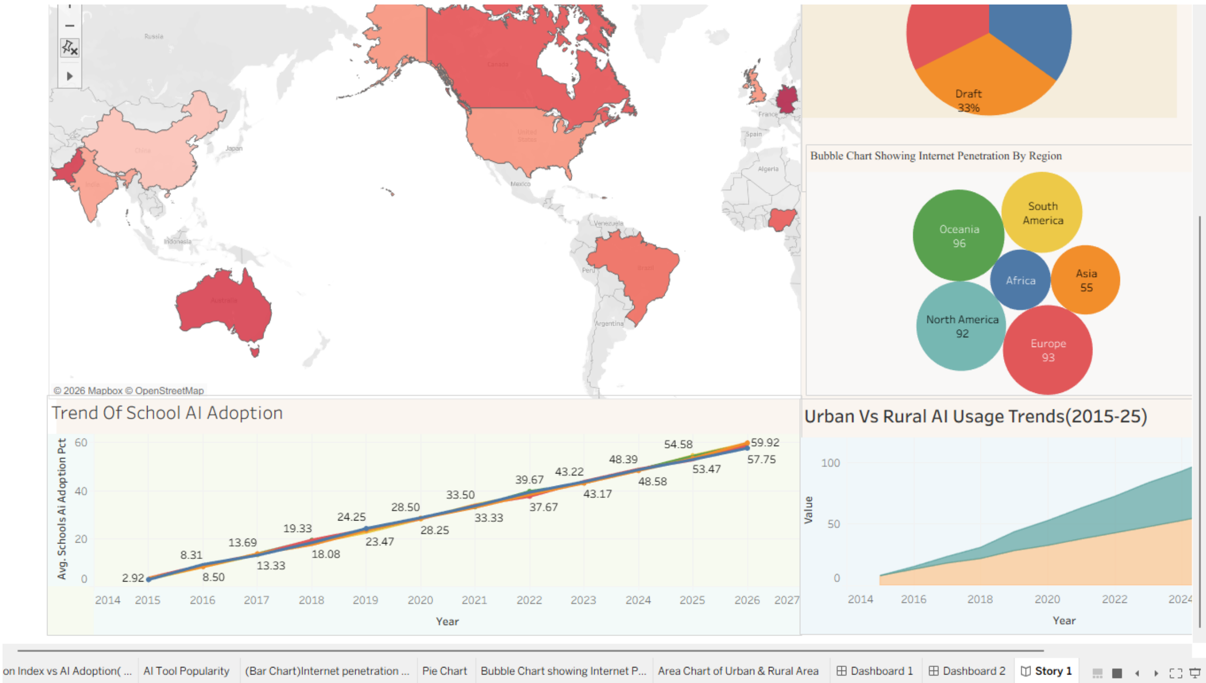
6.1 Story Design File

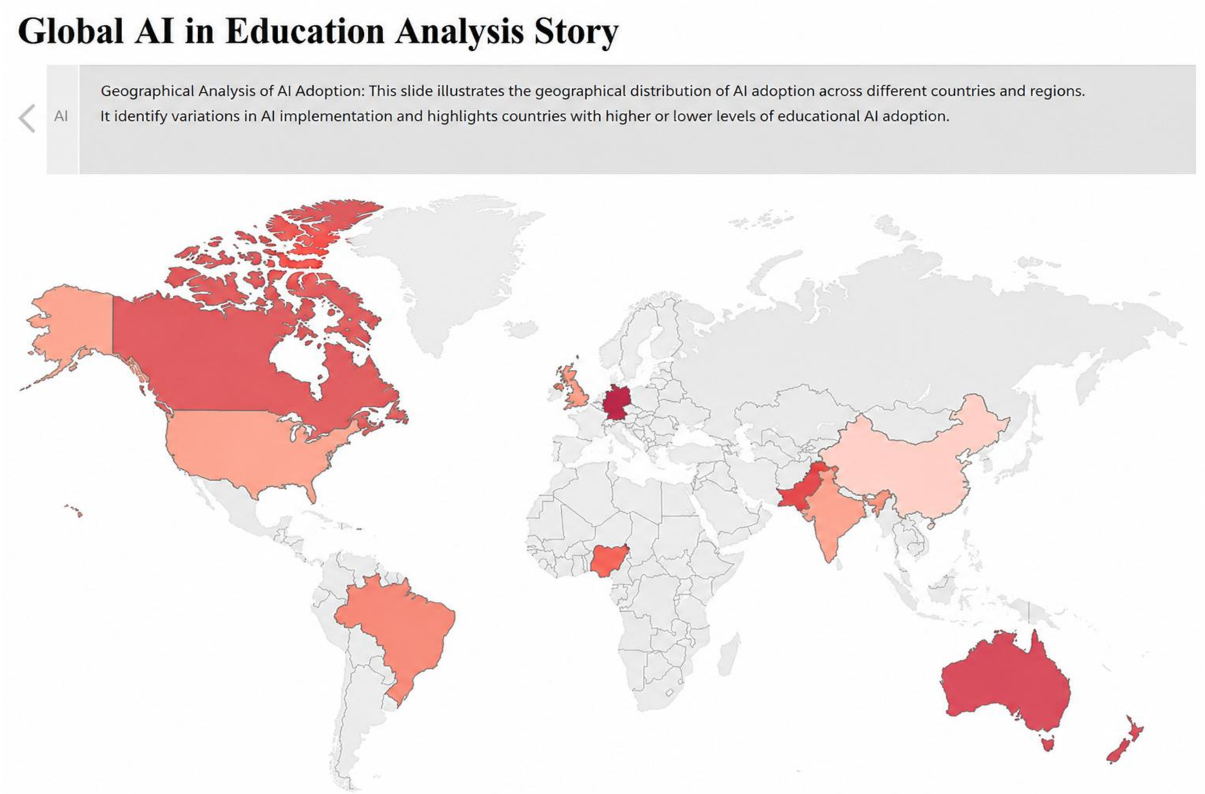
Story

Date	30 July 2026
Team ID	Shrishti Jaiswal
Project Name	Global AI in Education Analysis using Tableau
Maximum Marks	5 Marks

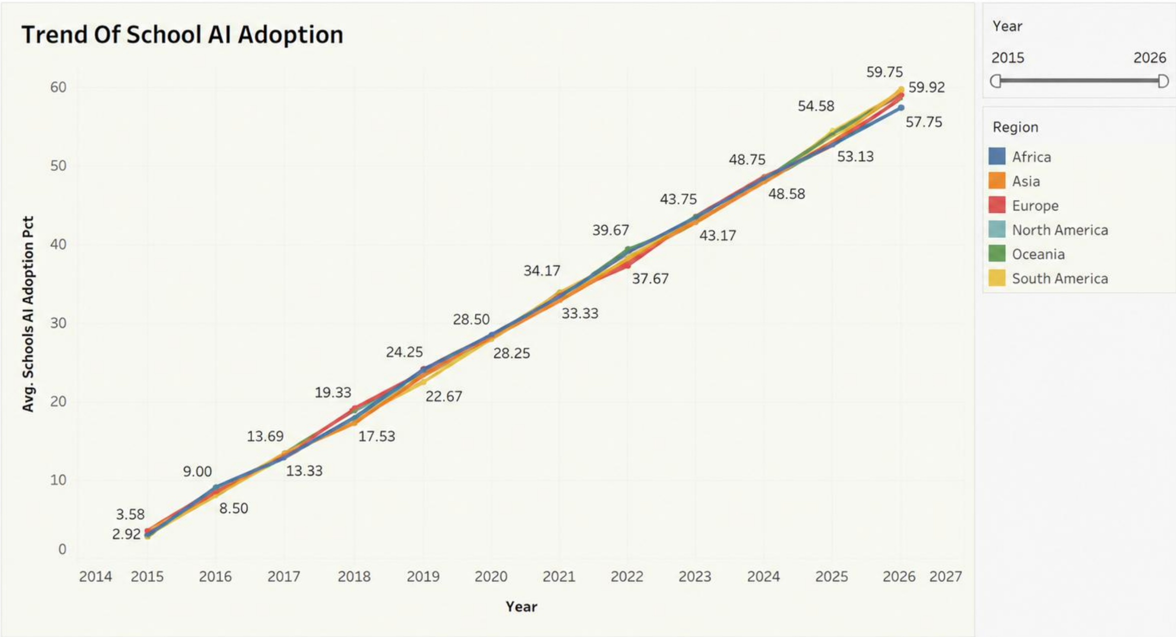
Activity 1: Story Design:




© 2026 Mapbox © OpenStreetMap
Draft
33%
Bubble Chart Showing Internet Penetration By Region
South America
Oceania
96
Asia
55
Africa
North America
92
Europe
93
Trend Of School AI Adoption
Avg. Schools AI Adoption Pct
Year
2.92
8.31
8.50
13.69
13.33
19.33
18.08
24.25
23.47
28.50
28.25
33.50
33.33
39.67
37.67
43.22
43.17
48.39
48.58
54.58
53.47
59.92
57.75
Urban Vs Rural AI Usage Trends(2015-25)
Value
Year
on Index vs AI Adoption(...
AI Tool Popularity
(Bar Chart)Internet penetration ...
Pie Chart
Bubble Chart showing Internet P...
Area Chart of Urban & Rural Area
Dashboard 1
Dashboard 2
Story 1

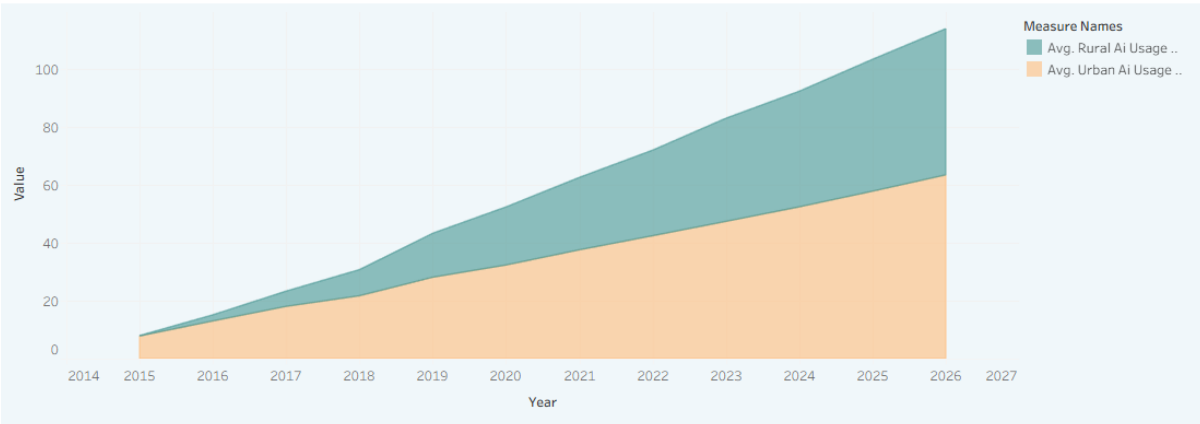

Global AI in Education Analysis Story
AI
Geographical Analysis of AI Adoption: This slide illustrates the geographical distribution of AI adoption across different countries and regions.
It identify variations in AI implementation and highlights countries with higher or lower levels of educational AI adoption.

School AI Adoption Trends (2015–2025):
This line chart illustrates the trend of AI adoption in schools over the years 2015 to 2025. It provides a clear understanding of how educational institutions gradually adopted AI technologies over time. The visualization helps identify periods of rapid growth or slower adoption, allowing educators and policymakers to evaluate long-term opportunities in educational AI implementation.



Global AI in Education Analysis Story

Urban vs Rural AI Usage Trends:
This visualization compares AI usage between urban and rural areas over the selected time period. It highlights differences in adoption levels, demonstrating how access to technology and digital infrastructure can influence AI implementation. The comparison provides valuable insights into the digital divide and emphasizes the importance of improving AI accessibilit across both urban and rural educational environments.



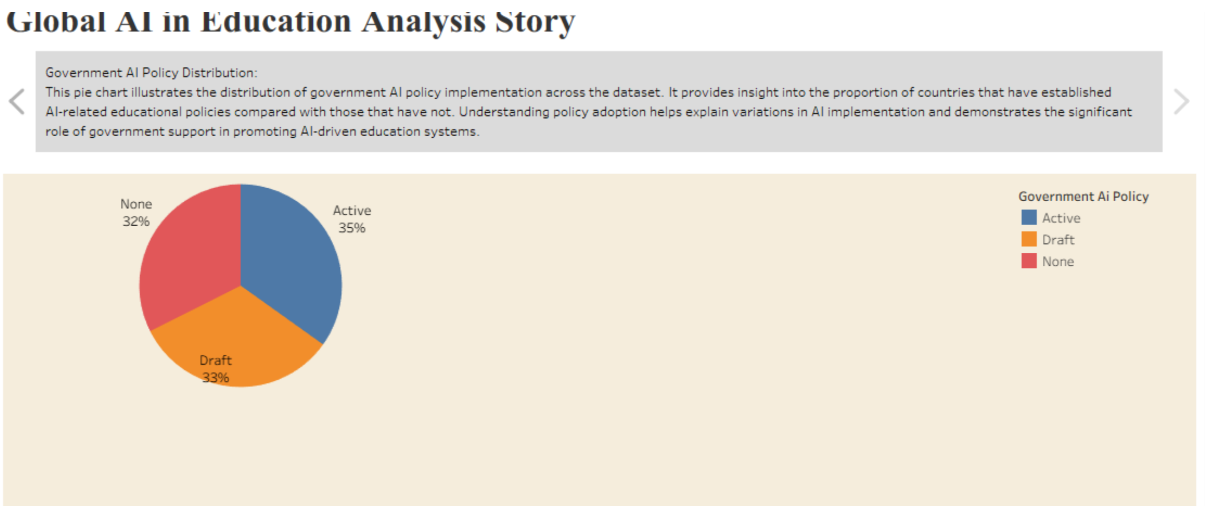


Tableau Story Narrative & Observations

By leveraging the sequential story point structure in Tableau, the **Global AI in Education Analysis** analytical presentation guides education stakeholders through a logical progression of AI adoption metrics, turning raw institutional data into a cohesive analytical narrative on the global state of AI in education.

Based on the interactive story points, here are the core analytical observations:

- **Adoption Has Accelerated Steadily Since 2015 (Adoption Trend):** The average share of schools adopting AI technologies climbed from under 3% in 2015 to nearly 60% by 2026, with Africa, Asia, Europe, North America, Oceania, and South America all tracking closely together throughout. This consistent, near-uniform growth curve suggests AI adoption in education is a global movement rather than a regionally isolated trend.
- **Geographic Concentration Among High-Adoption Nations (Geographical Distribution):** The world map view highlights Germany, Australia, and Canada as the countries with the deepest AI integration in education, while much of Africa, the Middle East, and large parts of Asia remain largely unshaded. This clear variation points to a persistent global gap in AI-driven learning infrastructure that mirrors broader digital-development divides.
- **Government Policy Still Catching Up (Policy Distribution):** Only 35% of countries in the dataset have an active government AI-in-education policy, while a combined 65% either have a policy still in draft (33%) or no policy at all (32%). This split shows that formal government support has not yet caught up with the pace of on-the-ground AI adoption.
- **A Persistent Urban-Rural Divide (Urban vs Rural Usage):** The area chart tracking AI usage from 2015 to 2026 shows urban schools consistently outpacing rural schools, with the gap between the two widening year over year rather than narrowing, evidence that the digital divide in education is intensifying rather than closing.

Activity 2:

Publish Story on Tableau Public and Paste Public link below

https://public.tableau.com/views/GlobalAIinEducationAnalysisusingTableau/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Web Deployment URL

https://public.tableau.com/views/GlobalAIinEducationAnalysisusingTableau/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Performance Testing

7.1 Utilization of Data Filters

Interactivity is one of the defining strengths of Tableau over static reporting, and this project makes deliberate use of filters so that every viewer can explore the dataset from their own perspective rather than being confined to a single fixed view. Two primary filters are used throughout the dashboards: a Year filter and a Region filter.

The Year filter is implemented as a slider spanning 2015 to 2026, matching the full time range covered by the dataset. It allows a viewer to move through the timeline and observe how metrics such as School AI Adoption Percentage or Urban and Rural AI Usage evolved year over year, rather than only seeing the final aggregated figure. This is particularly valuable for trend-based visualizations such as the multi-line adoption chart and the urban-versus-rural area chart, where the story lies as much in the trajectory as in the endpoint.

The Region filter allows the dashboard to be narrowed to any combination of Africa, Asia, Europe, North America, Oceania, and South America. On charts such as the School AI Adoption trend line and the Education Index scatter plot, this lets a user isolate a single region of interest, compare two or three regions side by side, or view all six at once, without needing separate dashboards for each geography.

Together, these filters convert what would otherwise be a set of static charts into a genuinely interactive analytical tool. They improve the user experience by letting viewers ask their own questions of the data, such as how one region's adoption compared to another's in a given year, or what the picture looked like earlier in the timeline, rather than relying solely on conclusions drawn by the report's author. This directly supports comparative analysis across time and geography, and ultimately supports better decision-making, since policymakers, administrators, and researchers can each filter the dashboard down to the specific slice of data most relevant to their own context before drawing conclusions.

7.2 Number of Calculation Fields

Calculation Field	Purpose
Average Daily AI Usage	KPI
School AI Adoption Percentage	KPI
Student AI Usage Percentage	KPI
Teacher AI Usage Percentage	KPI
Urban AI Usage Percentage	KPI
Rural AI Usage Percentage	KPI
Gender Gap in AI Usage	KPI

Calculation Field	Purpose
Average Internet Penetration	Bubble Chart
Average Education Index	Scatter Plot

Calculated fields are central to how this project turns a raw dataset into meaningful, decision-ready metrics. Rather than plotting raw columns directly, nine calculated fields were created in Tableau to aggregate and reshape the underlying data into the specific KPIs and analytical measures used throughout the dashboards and story. Seven of these, covering Average Daily AI Usage, School, Student, Teacher, Urban, and Rural AI Usage, and the Gender Gap in AI Usage, power the KPI cards that summarise the dataset at a glance. The remaining two support the more advanced analytical visualizations: Average Internet Penetration drives the packed bubble chart, while Average Education Index underpins the scatter-plot comparison against School AI Adoption. Without these calculated fields, each visualization would need to re-aggregate raw records individually, making the dashboards slower, less consistent, and harder to maintain; with them, every chart in the workbook draws on the same well-defined, reusable set of metrics.

7.3 Number of Visualizations

Visualization	Purpose
KPI Cards	Summary metrics
Filled Map	Country comparison
Scatter Plot	Correlation analysis
Line Chart	Trend analysis
Area Chart	Urban vs Rural comparison
Pie Chart	Policy distribution
Packed Bubble Chart	Regional internet penetration
Highlight Table	AI Tool Popularity
Horizontal Bar Chart	Internet penetration by country
Tableau Story	Sequential presentation

The ten visualizations listed above were not chosen arbitrarily; each was selected because its chart type is the most natural fit for the specific analytical question it answers. KPI cards were chosen because single summary numbers are best absorbed instantly, without requiring a viewer to interpret axes or colour scales, ideal for metrics like Average Daily AI Usage that stakeholders want to check at a glance. The Filled Map and the Horizontal Bar Chart both compare countries, but serve different purposes: the map emphasises geographic pattern and clustering, useful for spotting regional blocs of high or low adoption, while the bar chart is better suited to precise ranking, such as identifying which countries lead the group on internet penetration.

The Scatter Plot was the natural choice for testing a relationship between two continuous variables, Education Index and School AI Adoption, with bubble size adding a third dimension, Average Daily AI Usage, without needing an additional chart. Line and Area charts were both used for time-based comparisons, but for different structures: the Line Chart compares six separate regional trends most clearly when they are drawn as distinct lines, while the Area Chart is better suited to Urban versus Rural usage because the shaded regions make the size of the gap between the two categories immediately visible, not just its direction.

The Pie Chart was reserved for the one genuinely proportional, part-to-whole question in the analysis, how countries split across Active, Draft, and No government AI policy, a scenario pie charts handle well precisely because there are only three categories that sum to a meaningful whole. The Packed Bubble Chart compares six regions by internet penetration in a compact, visually engaging layout, while the Highlight Table was chosen for the AI Tool Popularity comparison because it needed to display a small grid of region-by-tool values where colour intensity, not shape, carries the comparative signal. Finally, the Tableau Story ties every one of these visualizations together into a single guided, sequential narrative, ensuring that a viewer encountering the workbook for the first time is led through the analysis in a logical order rather than left to interpret nine independent charts unaided.

8. Conclusion / Observation

The Global AI in Education Analysis project set out to convert a scattered set of indicators on AI adoption in education into a single, coherent, and interactive analytical resource, and the resulting dashboards and story support several clear observations about the current state of AI in global education.

The most consistent finding across every visualization is the sheer pace of global growth. Average School AI Adoption climbed from under 3 percent in 2015 to nearly 60 percent by 2026, and this growth was remarkably uniform across Africa, Asia, Europe, North America, Oceania, and South America alike, with all six regions tracking within a few percentage points of one another throughout the period. This near-uniform curve suggests that AI adoption in education is best understood as a genuinely global movement rather than a trend confined to a handful of wealthy regions.

At the same time, the country-level and infrastructure-level views reveal that this growth has not been evenly distributed. The Education Index versus AI Adoption scatter plot shows a positive association between a country's educational development and its level of AI adoption in schools, suggesting that stronger foundational education systems tend to create more favourable conditions for successful AI integration. Internet infrastructure tells a similar story: several of the higher-income countries in the dataset record internet penetration above ninety percent, placing them at the more digitally-ready end of the spectrum, while countries with considerably lower penetration face a steeper infrastructure barrier to AI-enabled learning.

Government policy readiness has not kept pace with this on-the-ground adoption. Only around 35 percent of countries in the dataset have an active government AI-in-education policy, while a combined 65 percent are still in draft, at 33 percent, or have no formal policy at all, at 32 percent. This gap between practice and policy is one of the more actionable findings of the project: schools and teachers are already adopting AI tools faster than governments are producing the regulatory frameworks to guide that adoption responsibly.

The urban-rural comparison provides the project's clearest cautionary observation. While AI usage has grown in both urban and rural areas every year since 2015, the area chart shows that urban usage has consistently outpaced rural usage, and the gap between the two has widened rather than narrowed over time. Left unaddressed, this suggests that the benefits of AI in education risk concentrating further in already better-connected urban areas rather than closing the digital divide.

Taken together, these findings demonstrate the value of building this analysis in Tableau rather than presenting it as a static report. Because every chart in the dashboard is filterable by year and region, the same workbook that shows global adoption climbing steadily also lets a policymaker isolate a single country or region, a researcher isolate a single time window, or an administrator compare their own region against the global average, all without rebuilding the analysis from scratch. This is the essence of business intelligence: not simply presenting data, but making it possible for different stakeholders to ask their own questions of the same underlying evidence. For governments, the dashboards highlight where policy is lagging behind practice; for researchers, they surface the correlation between education quality and AI readiness; for schools and universities, they provide a benchmark against which to measure their own digital transformation. In each case, the project's core observation stands: AI adoption in education is accelerating quickly and fairly globally, but infrastructure, policy, and equity have not yet caught up with that pace, and closing those gaps is the central challenge for the next phase of educational planning.

9. Future Scope

While the current project delivers a comprehensive snapshot of global AI adoption in education using the data available at the time of analysis, several extensions could meaningfully increase its depth, accuracy, and long-term value.

The most immediate opportunity is to move from a static, periodically-updated dataset to real-time educational data sources. Connecting the Tableau workbook to live feeds from ministries of education, international bodies, or institutional learning-management systems would allow the dashboards to refresh automatically rather than requiring manual data collection and cleaning for every update. Paired with Tableau's live-connection and scheduled-refresh capabilities, this would transform the current dashboards into genuinely live monitoring tools that stakeholders could consult at any time, rather than a fixed point-in-time analysis.

A second direction is the integration of machine learning and predictive analytics. The current project is descriptive, showing what has happened, but the same underlying dataset could support predictive models that forecast future AI adoption trajectories for individual countries or regions, flag countries at risk of falling further behind on infrastructure or policy, or estimate the likely impact of a given improvement in internet penetration on school AI adoption. Tools such as Tableau's native forecasting features, or external models connected through Python or R integration, could be layered onto the existing dashboards to add this predictive dimension without discarding the descriptive work already completed.

Student performance prediction represents a further extension, using classroom or institutional data, where available, to explore whether AI usage intensity correlates with measurable learning outcomes, rather than adoption alone. Similarly, institutional benchmarking tools could be built on top of the existing KPI framework, allowing an individual school or university to compare its own AI adoption profile against regional and global peers in a standardised way, turning the dashboard from a global overview into a practical self-assessment tool for individual institutions.

The scope of the dataset itself could also be expanded. The current analysis covers ten countries across six regions; broadening this to a larger and more representative set of countries, and adding further educational indicators such as teacher training levels, funding per student, or curriculum digitisation, would strengthen the statistical basis for the correlations already observed between Education Index, infrastructure, and AI adoption.

Finally, cloud deployment and cross-platform integration would improve the accessibility and reach of the work. Publishing the workbook through a managed cloud environment rather than a public one would allow controlled, permission-based access for institutional stakeholders, while building an equivalent version in Power BI or another business-intelligence platform would let the analysis reach organisations standardised on different tooling. None of these extensions would require abandoning the analytical foundation already built; each one adds a further layer, live data, prediction, broader scope, or wider accessibility, on top of the KPIs, dashboards, and story developed in this project, moving it from a one-time academic analysis toward a sustained, evolving decision-support tool for global education stakeholders.

10. Appendix

10.1 Source Code (if any)

No traditional programming source code was required because this project was developed entirely using Tableau Desktop/Public and Microsoft Excel. Tableau calculated fields, filters, parameters, dashboards, and stories were used for implementation.

10.2 GitHub & Project Demo

GitHub Repository: https://github.com/shrishtij2006/Global-AI-in-Education-Analysis-using-Tableau.git

Tableau Public Dashboard:

Dasboard 1:

https://public.tableau.com/views/GlobalAIinEducationAnalysisusingTableau/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dashboard 2 : https://public.tableau.com/views/GlobalAIinEducationAnalysisusingTableau/Dashboard2?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Project Demonstration Video: _____ https://drive.google.com/file/d/10mXOfXWpe7T-gZYQv10vmyy_DeRuf8I0/view?usp=sharing